

The Portfolio Management process also performs ongoing evaluation of the projects and systems in its portfolio. Based on periodic assessments, projects are determined to justify continued investment if they have the following characteristics:

- Contribute to the organization strategy
- Progress toward achieving established goals
- Comply with project directives from the organization
- Are conducted according to an approved plan
- Provide a service or product that is still needed and providing acceptable investment returns

Otherwise, projects may be redirected or, in extreme instances, terminated.

Inputs/Outputs Inputs and outputs for the Portfolio Management process are listed in [Figure 2.17](#). Descriptions of each input and output are provided in Appendix E.

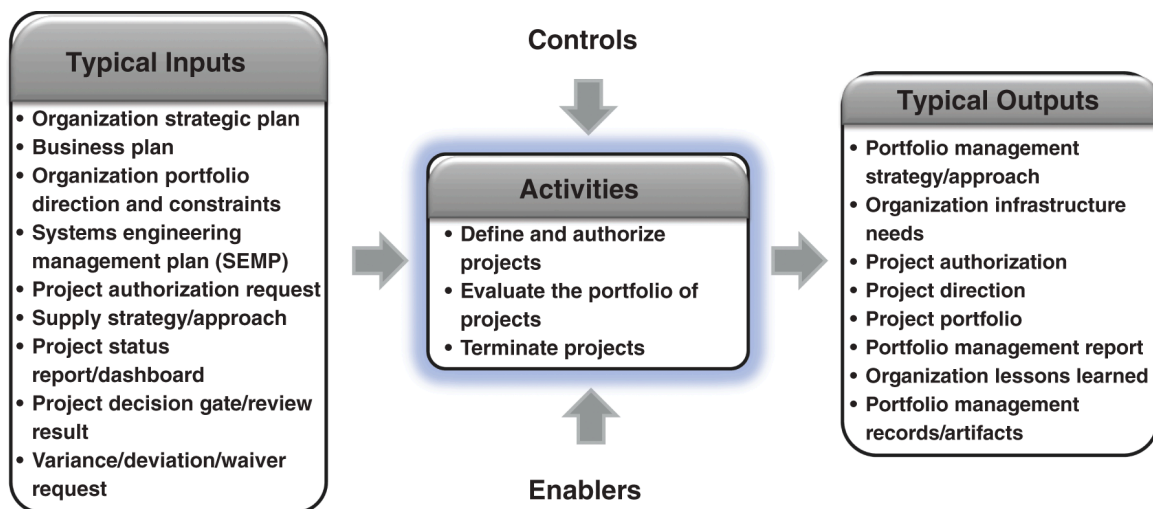


FIGURE 2.17 IPO diagram for Portfolio Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Portfolio Management process includes the following activities:

- *Define and authorize projects.*
 - – Obtain business area plans and organization strategic plans—use the strategic objectives to identify candidate projects to fulfill them.

- – Identify, assess, prioritize, and select investment opportunities consistent with the organization strategic plan.
- – Establish project scope, define project management accountabilities and authorities, and identify expected project outcomes.
- – Establish the domain area of the product line defined by its main features and their suitable variability.
- – Allocate adequate funding and other resources to selected projects.
- – Identify interfaces and opportunities for multi-project synergies.
- – Specify the project governance process including organizational status reporting and reviews.
- – Authorize project execution.
- *Evaluate the portfolio of projects.*
 - – Evaluate ongoing projects to provide rationale for continuation, redirection, or termination.
 - – Provide direction and supporting actions for continuation or redirection, as applicable for successful completion.
- *Terminate projects.*
 - – Close, cancel, or suspend projects that are completed or designated for termination.

Common approaches and tips:

- Logic modeling techniques that capture how an organization works can aid development or evaluation of business area plans at multiple levels of interest, ranging from the project- to portfolio-level plans (see, for example, PMBOK® (2021) Section 4.2 for a list of commonly used models). The logic models typically describe the fundamental theory/assumptions, planned work (resources, inputs and activities) linked with intended results (outputs, outcomes, and impact).
- When investment opportunities present themselves, prioritize them based on measurable criteria such that projects can be objectively evaluated against a threshold of acceptable performance. This assessment is done in the context of the business area planning to focus resources to best meet present and future objectives.

- Expected project outcomes should be based on clearly defined, measurable criteria to ensure that an objective assessment of progress can be determined. Specify the investment information that will be assessed for each milestone. Initiation should be a formal milestone that does not occur until all resources are in place as identified in the project plan.
- Establish organizational coordination mechanisms to manage the synergies between active projects in the organization portfolio. Complex and large organization architectures require the management and coordination of multiple interfaces and make additional demands on investment decisions. These interactions occur within and between the projects.
- Use a product line engineering approach (see [Section 4.2.4](#)) when stakeholders need the same or similar systems (e.g., common features), with some customizations (e.g., variants). The goal is to manage a product line as one product definition with planned variants as opposed to multiple separate products managed individually, thereby streamlining and simplifying the management effort.
- Include risk assessments (see [Section 2.3.4.4](#)) in the evaluation of ongoing projects. Projects that contain risks that may pose a challenge in the future might require redirection. Cancel or suspend projects whose disadvantages or risks to the organization outweigh the investment.
- Include opportunity assessments (see [Section 2.3.4.4](#)) in the evaluation of ongoing projects. Addressing project challenges may represent a positive investment opportunity for the organization. Avoid pursuing opportunities that are inconsistent with the capabilities of the organization and its strategic goals and objectives or contain unacceptably high technical risk, resource demands, or uncertainty.
- Allocate resources based on the requirements of the projects; otherwise, the risk of cost and schedule overruns may have a negative impact on quality and performance of the project.
- Establish effective governance processes that directly support investment decision making and communications with project management.

Elaboration

Define the Business Cases and Assess Against Business Area Plans. Portfolio management tries to maximize the benefit obtained by the organization from the use of financial assets and other resources within the organization. Thus, business cases for potential projects are evaluated for cost-benefit and the business need before a project is approved for the proposed SoI. Each decision gate reviews the business case as the project matures. The result is reverification or perhaps restatement of the business case.

The business case may be validated in a variety of ways. For large projects, sophisticated engineering models, or even prototypes of key system elements, help prove that the objectives of the business case can be met, and that the system will work as envisioned prior to committing large amounts of resources to full-scale engineering and manufacturing development. For smaller projects, when the total investment is modest, proof-of-concept models may be constructed during the concept stage to prove the validity of business case assumptions.

Investment opportunities are not all equal, and organizations are limited in the number of projects that can be conducted concurrently. Further, some investments are not well aligned with the overall strategic plan of the organization. For these reasons, opportunities are evaluated against the portfolio of existing agreements and ongoing projects, taking into consideration the attainability of the stakeholders' requirements.

Project Management and SE considerations. Portfolios may have multiple projects. As previously stated, projects are added to the portfolio after the candidate project can show that it is both feasible and meets organizational business needs. In many organizations projects with defined scope are organized in programs focused on a set of objectives that are part of the organization's strategic plan. As stated in the PMI (2017), the focus of portfolio management is "doing the right work" as opposed to program or project management which is more concerned with "doing work right."

The disciplines of project management and SE have overlapping responsibilities regarding portfolio management. To save time, share knowledge, facilitate the accomplishment of shared objectives, and achieve success, a strong partnership should exist between each of these disciplines (see [Section 5.3.3](#)).

At the portfolio level, the scope is extensive with consideration external to the organization and internal across the organization's enterprise. At the other end of the spectrum, the focus is internal to the project with consideration for the context of the product/service/result. An example of this is to look at the range in scope in requirements development, as shown in [Figure 2.18](#).

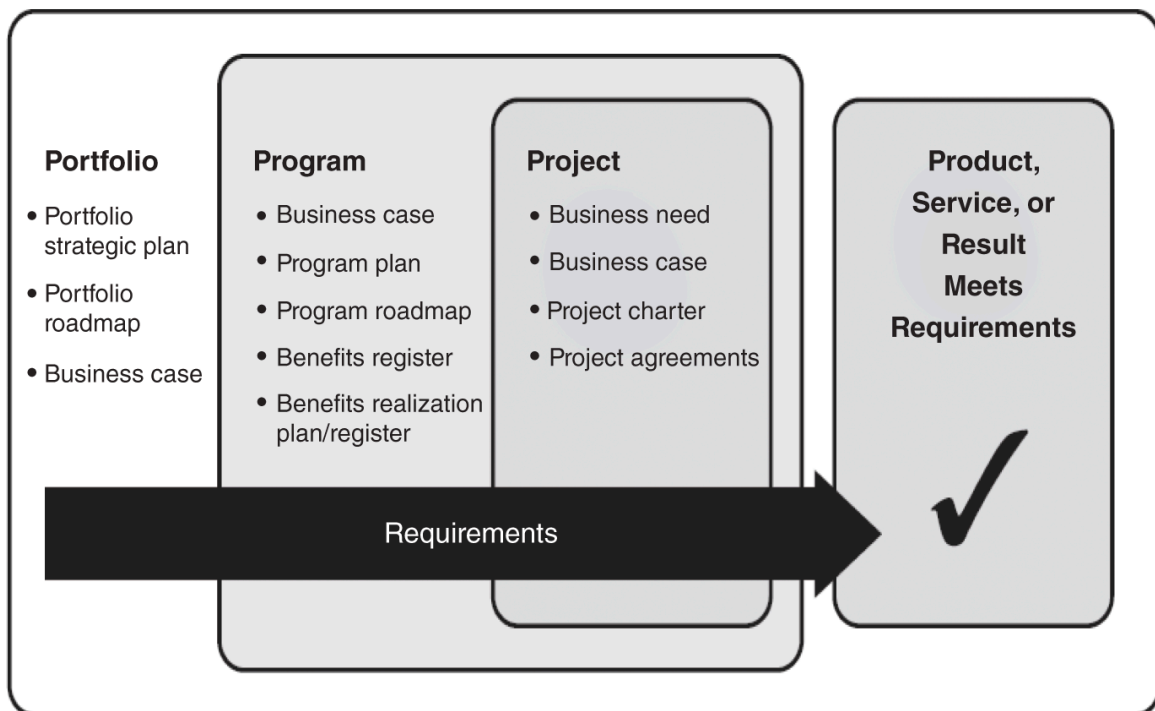


FIGURE 2.18 Requirements across the portfolio, program, and project domains. From PMI (2016). Used with permission. All other rights reserved.

At the portfolio level, the portfolio's strategic plan and roadmap address business and mission needs and provides direction and organizational focus, and plans/actions to realize the direction. Requirements often start at the concept or portfolio level as a high-level view associated with investment or business opportunities.

2.3.3.4 Human Resource Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.2.4.1] The purpose of the Human Resource Management process is to provide the organization with necessary human resources and to maintain their competencies, consistent with strategic needs.

Description Projects all need resources to meet their objectives. This process deals with human resources. Nonhuman resources, including tools, databases, communication systems, financial systems, and information technology, are addressed using the Infrastructure Management process (see [Section 2.3.3.2](#)).

Project planners determine the resources needed for the project by anticipating both current and future needs. The Human Resource Management process provides the mechanisms whereby the organization management is made aware of project needs and personnel are scheduled to be in place when requested. While this can be simply stated, it is less simply executed. Conflicts must be resolved, personnel must be trained, and employees are entitled to vacations and time away from the job.

The human resource management organization collects the needs, negotiates to remove conflicts, and is responsible for providing the personnel, without which nothing else can be accomplished. Since qualified personnel are not free, their costs are also factored into investment decisions.

Inputs/Outputs Inputs and outputs for the Human Resource Management process are listed in [Figure 2.19](#). Descriptions of each input and output are provided in Appendix E.

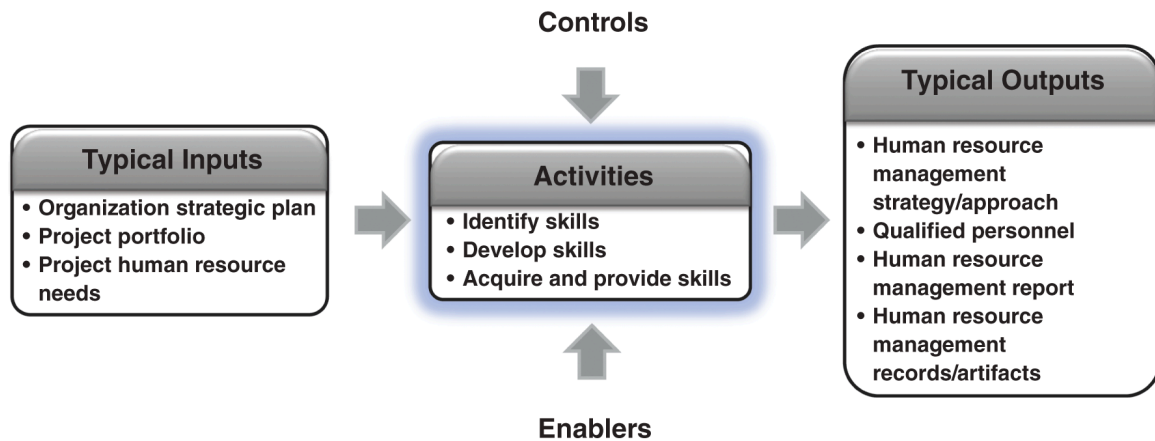


FIGURE 2.19 IPO diagram for Human Resource Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Processes Activities The Human Resource Management process includes the following activities:

- *Identify skills.*
 - – Identify and record skills of existing personnel to establish a “skills inventory.”
 - – Review current and anticipated projects to determine and record the skill needs across the portfolio of projects. The INCOSE Systems Engineering Competency Framework (SECF) (2018) and Systems Engineering Competency Assessment Guide (SECAG) (2023) can be used as resources to identify SE skills.
 - Evaluate skill needs against available personnel with the prerequisite skills to determine if training, hiring, or other skill acquisition activities are indicated.
- *Develop skills.*
 - – Establish a strategy/approach for skills development.
 - – Plan for the skill development per the strategy.
 - Obtain (or develop) and deliver training, education, and mentoring to close identified gaps of project personnel.
 - – Identify skills, abilities, and behaviors needed for competencies. The INCOSE Systems Engineering Assessment Guide is a recommended resource for this.
 - – Identify training and development resources to match desired skills, abilities, and behaviors development. The INCOSE Professional Development Portal can help identify potential resources.

- – Identify assignments that lead toward career progression.
- – Create succession plans to ensure that the desired skill set and flow of skill development through the organization is sustained into the future.
- – Create and maintain skill development records.
- *Acquire and provide skills.*
 - – Provide human resources to support all projects.
 - – Train or hire qualified personnel when gaps indicate that skill needs cannot be met with existing personnel.
 - – Maintain and manage a skilled personnel pool to staff ongoing projects.
 - – Assign personnel to projects based on personnel development and project needs.
 - – Create and maintain staff assignment records.
 - – Motivate personnel by providing career development and reward programs.
 - – Resolve personnel conflicts between or within projects.
 - – Maintain communication across projects to effectively allocate human resources throughout the organization and identify potential future or existing conflicts and problems with recommendations for resolution.
 - – Schedule other related assets or, if necessary, acquire them.
- *Develop and Manage Competencies.*
 - – Create and maintain job role definitions related to competencies required.
 - – Identify organization competency gaps.
 - – Align organization competencies with strategic objectives.
 - – Maintain organization-level competency definitions and frameworks.

Common approaches and tips:

- The availability and suitability of personnel is one of the critical project assessments and provides feedback for improvement and reward mechanisms.

- Consider using an IPDT environment as a means to reduce the frequency of project rotation, recognize progress and accomplishments and reward success, and establish apprentice and mentoring programs for newly hired employees and students.
- Maintain both a listing of skill needs and the paths to obtain the necessary expertise, including a pipeline of candidates, training provisions, consultants, temporary outsourcing, reassignments, etc.
- Personnel are allocated based on requests and conflicts are negotiated. The goal is to provide personnel to a project when they are needed to keep the project on target and on budget.
- Try to avoid the overcommitment of project personnel, especially people with specialized skills.
- Skills inventory and career development plans are important documentation that can be validated by engineering and project management. The INCOSE SECF and SECAG are comprehensive resources of skills that can be used to develop career development plans.
- Maintain an organization career development program that is not sidetracked by project demands. Develop a policy that all personnel receive training or educational benefits on a regular cycle. This includes both undergraduate and graduate studies, in-house training courses, certifications, tutorials, workshops, and conferences.
- Remember to provide training on organization policies and procedures and system life cycle processes.
- Establish a resource management information infrastructure with enabling support systems and services to maintain, track, allocate, and improve the resources for present and future organization needs.
- Use the slack time in the beginning of a project to provide training to ensure necessary skills.
- Career development plans should be managed and aligned to the objectives of both the employee and the organization. Career development plans should be reviewed, tracked, and refined to provide a mechanism to help manage the employee's career within the organization.

Elaboration

Human Resource Management Concepts. The Human Resource Management process maintains and manages the people required by the portfolio of organization projects. Human resource management is the efficient and effective deployment of qualified personnel when and where they are needed. A balance should be found between efficiency and robustness. Human resource management relies heavily on forecasts into the future of the demand and supply of various resources.

The primary objective of this process is to provide a pool of qualified personnel to the organization. This is complicated by the number of sources for requests, the need to balance the skills of the labor pool against the other infrastructure elements (e.g., computer-based tools), the need to maintain a balance between the budgets of individual projects and the cost of resources, the need to keep apprised of new or modified policies and procedures that might influence the skills inventory, and myriad unknowns.

Project managers face their resource challenges competing for scarce talent in the larger organization pool. They must balance access to the experts they need for special studies with stability in the project team with its tacit knowledge and project memory. Today's projects depend on teamwork and optimally multidisciplinary teams. Such teams are able to resolve project issues quickly through direct communication between team members. Such intrateam communication shortens the decision-making cycle and is more likely to result in improved decisions because the multidisciplinary perspectives are captured early in the process.

2.3.3.5 Quality Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.2.5.1] The purpose of the Quality Management process is to assure that products, services, and implementations of the Quality Management process meet organizational and project quality objectives and achieve customer satisfaction.

Description The overarching process for achieving quality goals is the Quality Management (QM) process and its supporting methods, values, and subprocesses. Properly communicated, through policy and procedure, it makes visible the goals of the organization to achieve customer satisfaction. These goals, when supported by measurable activities, provide feedback for maintaining consistency in work processes and delivering quality outcomes. Since primary drivers in any project are time, cost, and quality, inclusion of a comprehensive QM process and its subprocesses is essential to every organization and must be sustained by a work culture that is disciplined in the proper execution of QM foundational principles and values. System life cycle processes are concerned with quality issues, and this is sufficient justification for spending the time, money, and energy into establishing QM fundamentals in an organization, its processes, and its people.

The QM process for SE ensures that all SE processes are deployed consistently by capable staff that can then produce systems designs that fulfill the stakeholder's requirements and lead to development and build processes that are aligned to produce high levels of performance throughout the organization.

Inputs/Outputs Inputs and outputs for the Quality Management process are listed in [Figure 2.20](#). Descriptions of each input and output are provided in Appendix E.

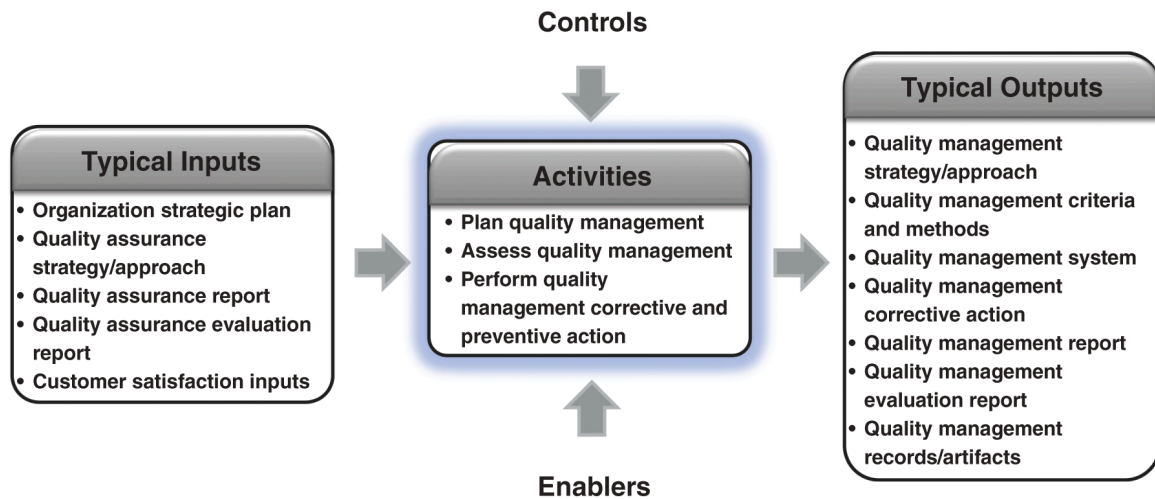


FIGURE 2.20 IPO diagram for the Quality Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Quality Management process includes the following activities:

- *Plan quality management.*
 - – Identify, assess, and prioritize quality guidelines consistent with the organization strategic plan. Establish QM guidelines-policies, standards, and procedures.
 - – Establish organization and project QM goals and objectives, including QM Culture emphasis.
 - – Establish organization and project QM responsibilities and authorities.
- *Assess quality management.*
 - – Evaluate project assessments.
 - – Assess customer satisfaction against compliance with requirements and objectives.
 - – Continuously improve the QM guidelines.
- *Perform quality management corrective action and preventive action.*
 - – Recommend appropriate action, when indicated.
 - – Maintain open communications within the organization and with stakeholders.

Common Approaches and Tips

- Management's commitment to quality is reflected in the integration of QM principles in the strategic planning and budgeting of the organization, and the allocation of educational resources to achieve and sustain a reliable QM culture.
- A quality policy, mission, strategies, goals, and objectives provide essential inputs along with a description of an organization's fundamental values for supporting a growing QM culture.

Elaboration

QM Generally accepted theory and practice. The four generally accepted foundational values of quality are its definition, its system, the standard for quality, and the method for measuring quality. Philip Crosby called them the Four Absolutes of Quality (Crosby, 1979).

1. *The definition of quality is meeting the stakeholder's requirements, needs and expectations.* Organizations (and individuals) are both producers and users of systems. One organization or person (acting as an acquirer) can task another (acting as a supplier) for products or services. This transaction is achieved using agreements that promise to fulfil the stakeholder's requirements in exchange for something of value, usually money. Quality pioneer W. Edwards Deming stressed that meeting stakeholder needs represents the defining criterion for quality and that all members of an organization need to participate actively in "constant and continuous" quality improvement (Deming, 1986).
2. *The system of quality is prevention.* One of the two QM prevention methods is Quality Assurance (QA). QA can be described as "putting good things into our processes" so that they perform as designed and conform to our stakeholder's requirements. QA was born in the aerospace industry and was originally referred to as "reliability engineering." It is generally associated with activities such as failure testing and pre-inspecting batches of materials and system elements that are then certified for use, thus preventing errors and defects from occurring by building-in quality. The QA methodology also includes infusing processes with reliable human resources and appropriate

policies, procedures, and training (SEH [Section 2.3.4.8](#)). Quality Control (QC) is the QM method for “taking bad things out of our processes after they occur” to prevent the defects that are discovered from reaching our stakeholders. QC includes checking, monitoring, and inspecting for defects and the removal, replacement, or rework of defective outcomes. One method of monitoring and statistically evaluating the stability and potential defect rates of processes is Statistical Process Control (SPC). Many manufacturing and high-volume service organizations use SPC to help achieve quality. Traditional SPC techniques include real-time, random sampling to test a fraction of the output for variances within critical tolerances (Juran, 1974).

3. *The standard for Quality is Zero Defects (ZD)*. It is important to make a distinction between the tracking of defects from feedback loops to improve our processes and progress toward a ZD count, and the more fundamental human term which is the Zero Defects Attitude (ZDA) (Kennedy, 2005). A ZDA is not about achieving perfection; it is a commitment to make each stakeholder’s experience as close to what was promised as possible. No one can achieve perfection, nor attain and sustain ZD, so we cannot expect perfection from any of our staff or processes. Like the “pride of workmanship,” people with a ZDA have a “heart attitude” that desires to prevent all defects and to reach the highest level of personal performance and customer satisfaction. People with a ZDA want to keep their promises to everyone and make things right when we fail. A ZDA, coupled with appropriate metrics and plans to progress toward ZD, will result in continuous and incremental improvement.
4. *The method for measuring quality is the price of non-conformance* (Crosby, 1979). It is a calculation of the expenses incurred by defects and their related rework, replacement, warranties, customer service, etc. The American Society for Quality calls it the “cost of poor quality.” It is an essential factor in calculating the actual “cost of quality” which is determined by comparing the “price of non-conformance (or doing things wrong) that includes expense caused by re-work, defects, and warranties, with the “price of conformance (or doing things right)” which is a calculation of the expenses related to improving processes

and applying preventive methods. The cost of quality includes a calculation of quantitative and qualitative parameters that are measured in both financial and human values. When the cost of doing things right is equal to or less than the price of non-conformance then, as Crosby said, “Quality is Free.”

QM Culture. SE practitioners need to have sufficient process knowledge and a QM knowledge base to be able to evaluate prevention options and make continuous, incremental improvements. When engineering disciplines are supported by planning and budgeting skills that resonate with the organization, we can achieve Process Quality with effective, efficient, and profitable outcomes, low defect rates, and delighted stakeholders. Deming, in his “14 Points” emphasized the need to “create constancy of purpose for improving products and services” and that it should be supported by “a vigorous program of education and self-improvement for everyone” (Deming, 1986). A high-performing work culture is measured by identifiable attributes or values within an organization's leadership style and workforce that directly influence the reliability of outputs. Kennedy (2005) leverages Deming’s mandate and the work of Crosby by defining the Eight Attributes of a Quality Management Culture that are described in [Table 2.3](#). [Figure 2.21](#) shows a QM culture resulting from QM values and skills integration.

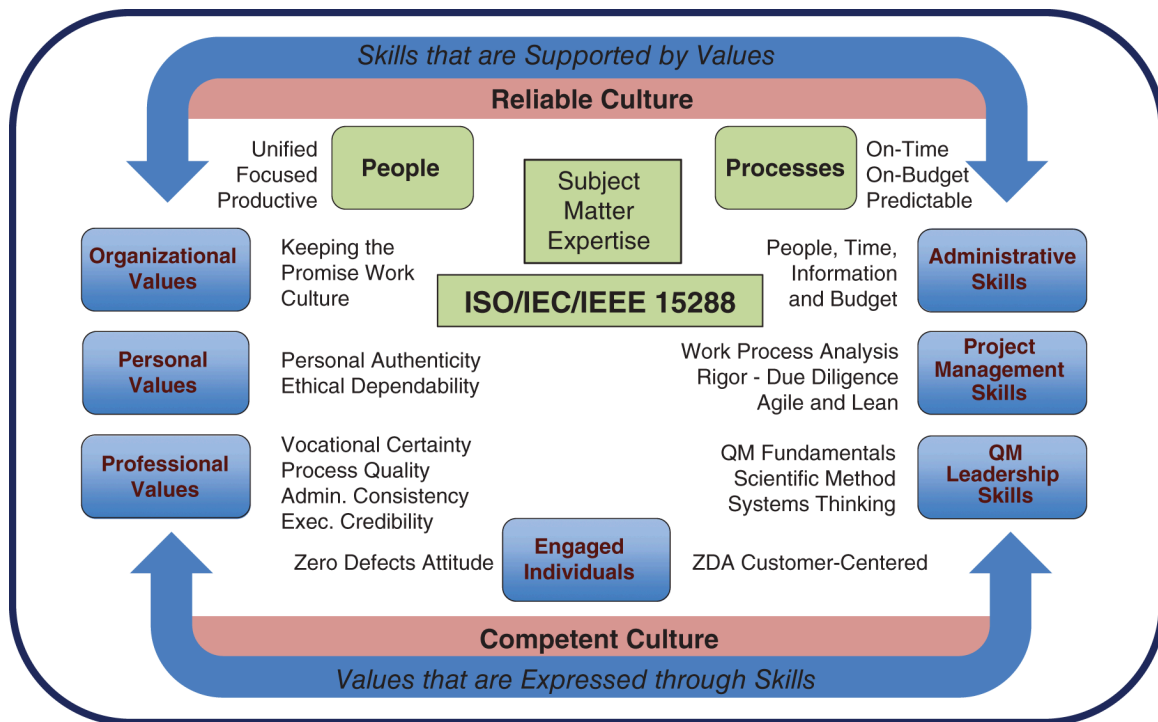


FIGURE 2.21 QM Values and Skills Integration. From Kennedy (2005). Used with permission. All other rights reserved.

TABLE 2.3 Eight Attributes of a Quality Management Culture

1. Zero Defects Attitude: A measure of our commitment to keep our promises and to initiate systems with the goal of preventing defects from reaching our customers.

2. Vocational Certainty: A measure of our faithfulness to our career agenda. A QM culture is disciplined about developing their skills and talents and acquiring earned confidence.

3. Process Quality: A measure of our mastery of planning and budgeting disciplines and how effectively we apply them to create viable work processes.

4. Administrative Consistency: A measure of our attention to details. QM cultures carefully listen to their customer to identify and conform to their requirements and assure customer satisfaction.

5. Executive Credibility: A measure of our sincerity and skill with people. Sincerity comes naturally from the heart, but skills can be sharpened and improved to gain reliable influence.

6. Personal Authenticity: A measure of our resolve to be consistent with our customers and coworkers. Authentic QM cultures work diligently to make exceptional service feel normal.

7. Ethical Dependability: A measure of our trustworthiness in practical matters. QM cultures are what we turn to when we want things to work right, run on time, and be there when needed.

8. Create a Keeping the Promise Culture: A measure of the mutual respect, accountability, and professionalism in a work culture. These are the practiced values of effective QM cultures.

From Kennedy (2005). Used with permission. All other rights reserved.

2.3.3.6 Knowledge Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.2.6.1] The purpose of the Knowledge Management process is to create the capability and assets that enable the organization to exploit opportunities to re-apply existing knowledge.

Description Knowledge Management (KM) includes the identification, capture, creation, representation, dissemination, and exchange of knowledge across targeted groups of stakeholders. It draws from the

insights and experiences of individuals and/or organizational groups or projects. The knowledge includes both explicit knowledge (conscious realization of the knowledge, often captured in artifacts and able to be communicated) and tacit knowledge (internalized in an individual or team without conscious realization) and can come from either individuals (through experience) or organizations (through processes, practices, and lessons learned) (Alavi and Leidner, 1999) (Roedler, 2010).

Within an organization, explicit knowledge is usually captured in its training, processes, practices, methods, policies, and procedures. In contrast, tacit knowledge is embodied in the individuals or teams of the organization and requires specialized techniques to identify and capture the knowledge, if it is to be passed along within the organization.

KM efforts typically focus on organizational objectives such as improved performance, competitive advantage, innovation, the sharing of good practices or lessons learned, avoidance of relearning practices, integration, and continuous improvement of the organization (Gupta and Sharma, 2004). KM captures knowledge that would otherwise be lost. So, it is generally advantageous for an organization to adopt a KM approach that includes building the framework, assets, and infrastructure to support the KM.

In this handbook, KM is viewed from an organizational project-enabling perspective, that is, how the organization supports the project (or program) environment with the resources in its KM system. The support provided to the project can come in several ways, including:

- Knowledge captured from technical experts.
- Lessons learned captured from previous similar projects.
- Domain engineering information that is applicable for reuse on the project, such as part of a product line or system family (see [Section 4.2.4](#)).
- Architecture or design patterns that are commonly encountered.
- Other reusable assets that may be applicable to the SoI.

Inputs/Outputs Inputs and outputs for the Knowledge Management process are listed in [Figure 2.22](#). Descriptions of each input and output are provided in Appendix E.

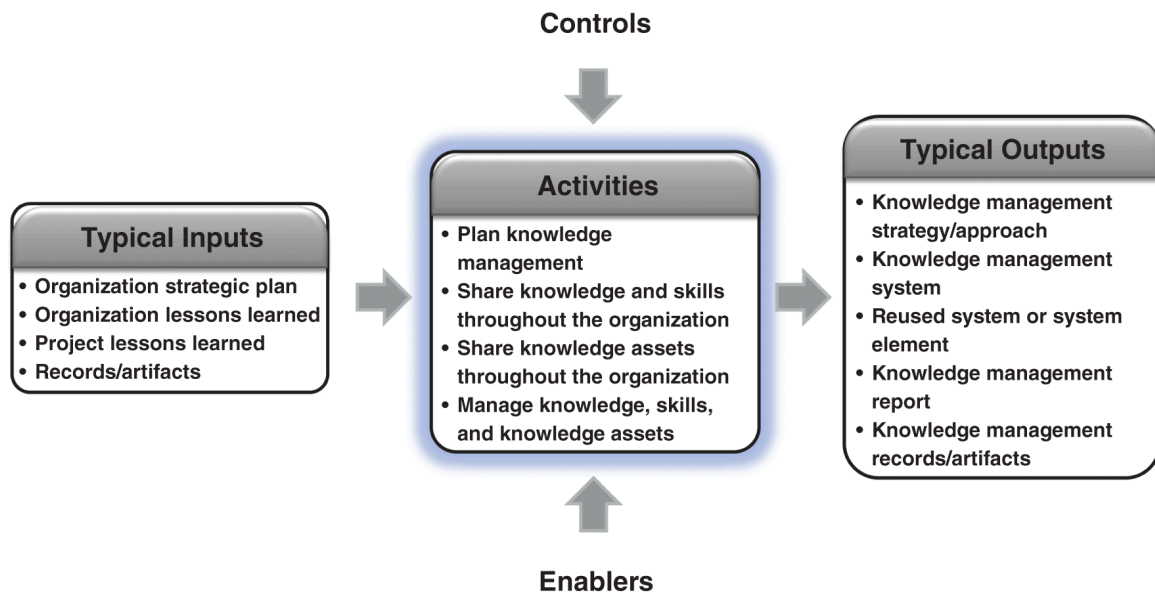


FIGURE 2.22 IPO diagram for Knowledge Management process. INCOSE original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The knowledge management process includes the following activities:

- *Plan knowledge management.*
 - – Establish a KM strategy that defines the approach and priorities for how the organization and projects within the organization will interact to ensure the right level of knowledge is captured to provide useful knowledge assets.
 - – Establish the scope of the KM strategy—the organization and projects need to identify the specific knowledge information to capture and manage. Considerations include the importance and cost effectiveness of capturing the knowledge. If there is no identified project that will benefit from the knowledge asset, then it probably should not be considered.
- *Share knowledge and skills throughout the organization.*
 - – Capture, maintain, and share knowledge and skills per the strategy. The infrastructure should be established to include mechanisms to

easily identify, access, and determine the applicability of the knowledge and skills.

- *Share knowledge assets throughout the organization.*
 - – Establish a taxonomy for the reapplication of knowledge.
 - – Establish a representation for domain models and domain architectures to help ensure an understanding of the domain and identify and manage opportunities for common system elements and their representations, such as architecture or design patterns, reference architectures, and common requirements.
 - – Define or acquire the knowledge assets applicable to the domain, including system and software elements, and share them across the organization. As the system and system elements are defined in the Technical Processes, the information items that represent those definitions should be captured and included as knowledge assets for the domain. The infrastructure should be established to include mechanisms to easily identify, access, and determine the applicability of the assets.
- *Manage knowledge, skills, and knowledge assets.*
 - – As the domain, family of systems, or product line changes, ensure the associated knowledge assets are revised or replaced to reflect the latest information. In addition, the associated domain models and architectures also may need to be revised.
 - – Assess and track where the knowledge assets are being used. This can help understand the utility of specific assets, as well as determine whether they are being applied where they are applicable.
 - – Determine whether the knowledge assets reflect current technology and continue to evolve.

Common approaches and tips:

- The planning for KM may include:
 - – Plans for obtaining and maintaining knowledge assets for their useful life.
 - – Characterization of the types of assets to be collected and maintained along with a scheme to classify them.

- – Criteria for accepting, qualifying, and retiring knowledge assets.
- – Procedures for controlling changes to the knowledge assets.
- – A mechanism for knowledge asset storage and retrieval.
- In developing an understanding of the domain, it is important to identify and manage both the commonalities (such as features, capabilities, or functions) and the differences or variations of the system elements (including where a common system element has variations in parameters depending on the system instance). The domain representations should include:
 - – Definition of the boundaries.
 - – Relationships of the domains to other domains.
 - – Domain models that incorporate the commonalities and differences allowing for sensitivity analysis.
 - – An architecture for a system family or product line within the domain, including their commonalities and variations (see [Section 4.2.4](#)).

Elaboration

General KM Implementation. KM focuses on capturing the organizational, project, and individual knowledge for use throughout the organization in the future. It is important to capture end-of-project lessons learned prior to the project personnel moving on to new assignment. However, an effective Knowledge Management process has the knowledge capture mechanisms in place to capture the relevant information throughout the life of the project, rather than trying to piece it together at the end.

KM for Product Lines and Reuse. KM also includes identification of systems that are part of a product line or system family (see [Section 4.2.4](#)) and system elements that are designed for reuse. For the first instance of these systems and system elements, the KM system needs to capture the domain engineering artifacts in a way to facilitate their use in the future. For subsequent instances, the KM system needs to provide the domain engineering information and capture any variations, updates of technology, and lessons learned. Issues important to the organization include:

- Definition and planning of KM activities for domain engineering and asset preservation, including tasks dedicated to domain engineering of product lines or system families and to the preservation of reusable assets.
- Integration of architecture management into the KM system including frameworks, architecture reuse, architecture reference models, architecture patterns, platform-based engineering, and product line architecture.
- Characterization of the types of assets to be collected and maintained including an effective means for users to find the applicable assets.
- Determination of the quality and validity of the assets.

Potential Reuse Issues. There are serious traps in reuse, especially with respect to commercial off-the-shelf (COTS) (see [Section 4.3.3](#)) and non-developmental item (NDI) elements:

- Do the new system or system element requirements and operational characteristics closely match the prior one? Trap: the prior solution was intended for a different use, environment, or performance level, or it was only a prototype.
- How did the prior system or system element perform? Trap: it worked perfectly, but the new application is outside the qualified range (e.g., using a standard car for a high-speed track race).
- Is the new system or system element going to operate in the same environment as the prior one? Trap: it is not certain, but there is no time to study it. One NASA Mars probe was lost because the development team used a radiator design exactly as was used on a successful satellite in Earth orbit. When the Mars mission failed, the team then realized that Earth orbiting environment, while in space, is different from a deep space mission.
- Is the system/system element definition defined and understood (i.e., requirements, constraints, operating scenarios, etc.)? Trap: too often, the development team assumes that if a reuse solution will be applied (especially for COTS), there is no need for well-defined system definition.

The issues may not show up until systems integration, causing major cost and schedule perturbations.

- Is the solution likely to have emergent requirements/behaviors where the reuse is being considered? Trap: a solution that worked in the past was used without consideration for the evolution of the solution. If COTS is used, there may be no way to adapt or modify it for emergent requirements.

A properly functioning KM system paired with well-defined processes and engineering discipline can help avoid these problems.

2.3.4 Technical Management Processes

The engineering of new or existing systems is managed by the conduct of projects. For this reason, it is important to understand the contribution of SE to the management of the project. This contribution is provided through the Technical Management Processes, which ensure the successful management of the SE effort within the project.

The Technical Management Processes are defined in ISO/IEC/IEEE 15288 as follows:

[5.7.4] The Technical Management Processes are concerned with managing the resources and assets allocated by organization management and with applying them to fulfill the agreements into which the organization or organizations enter. The Technical Management Processes relate to the technical effort of projects, in particular to planning in terms of cost, timescales and achievements, to the checking of actions to help ensure that they comply with plans and performance criteria, and to the identification and selection of corrective actions that recover shortfalls in progress and achievement. They are used to establish and perform technical plans for the project, manage information across the technical team, assess technical progress against the plans for the system products or services, control technical tasks through to completion, and to aid in the decision-making process.

Technical management, which is the application of technical and administrative resources to plan, organize and control engineering functions, consists of the following eight processes: Project Planning, Project Assessment and Control, Decision Management, Risk Management, Configuration Management, Information Management, Measurement, and Quality Assurance. The Technical Management Processes are used consistently throughout the system life cycle so that system-specific Technical Processes can be conducted effectively. They work with the project management processes to establish and perform technical plans, manage information across the technical teams, assess technical progress against the plans, control technical tasks and risks through to completion, and aid in the decision-making process.

SE practitioners continually interact with project management practitioners. Both contribute to the project with unique professional competences. A life cycle from the project management practitioner's point of view (project start–project end) is defined differently than from the SE practitioner's point of view (system concept to system retirement). But there is a “shared space” where both must collaborate to drive the team's

performance and success (Langley, et al., 2011). See [Section 5.3.3](#) for treatment of the integration between SE and project management.

2.3.4.1 Project Planning Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.3.1.1] The purpose of the Project Planning process is to produce and coordinate effective and workable plans.

Description Project planning starts with the identification of a new potential project and continues after the authorization and activation of the project until its termination. The Project Planning process is performed in the context of the organization, and in compliance with the Life Cycle Model Management process (see [Section 2.3.3.1](#)) that identifies and establishes relevant policies and procedures applicable to all projects owned by the organization.

The Project Planning process identifies the project objectives, technical activities, interdependencies, resource requirements, risks and opportunities, and management approach for the technical effort. The planning includes the estimates of needed resources and budgets and the determination of the need for project enablers, including specialized equipment, facilities, and specialists during the project to improve efficiency and effectiveness and decrease cost overruns. This requires coordination across the set of processes to develop a set of consistent planning for all activities. For example, different disciplines work together in the performance of the System Requirements Definition, System Architecture Definition, and Design Definition processes to evaluate the parameters such as producibility, testability, operability, maintainability, and sustainability against product performance. Project tasking may be concurrent to achieve the best results.

Project planning establishes the direction necessary to enable execution of the project and the assessment and control of the project progress. It identifies the details of the work and the right set of personnel, skills, infrastructure, and facilities with a schedule for needed resources from within and outside the organization.

Inputs/Outputs Inputs and outputs for the Project Planning process are listed in [Figure 2.23](#). Descriptions of each input and output are provided in Appendix E.

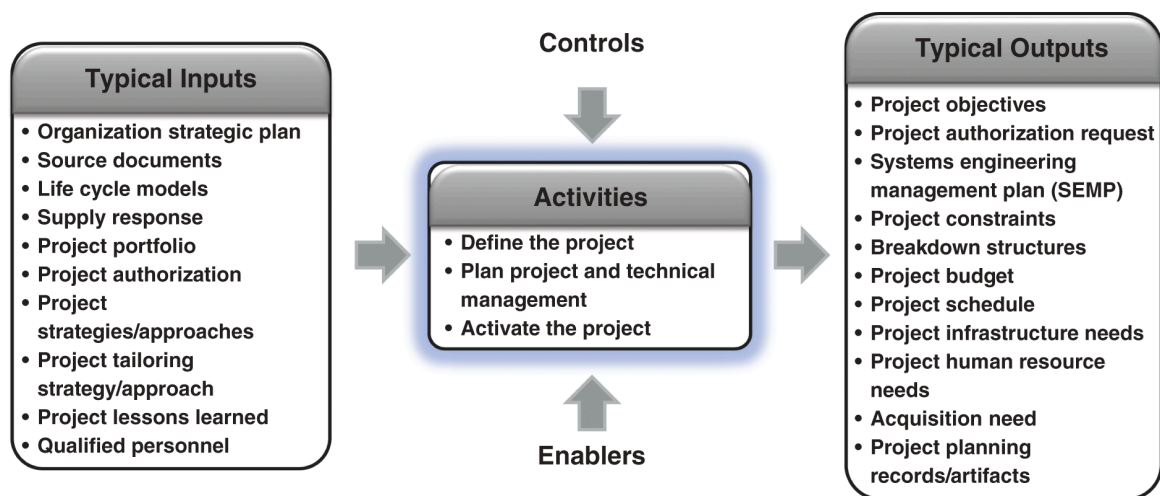


FIGURE 2.23 IPO diagram for Project Planning process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Project Planning process includes the following activities:

- *Define the project.*
 - – Analyze the project supply response and related agreements to define the project objectives, assumptions, constraints, and scope.
 - – Identify or establish tailoring of organization procedures and practices to carry out planned effort (see [Section 4.1](#)).
 - – Develop or select Breakdown Structures based on the evolving system architecture (see paragraph on Breakdown Structures hereafter) and the constraints on the resources.
 - – Define and maintain a life cycle model that could be tailored from the defined life cycle models of the organization. This includes the

identification of major milestones, decision gates, and project reviews.

- *Plan project and technical management.*
 - – Establish the roles and responsibilities for project authority.
 - – Define top-level work packages for each activity identified. Each work package should be tied to required resources including procurement strategies.
 - – Develop a project schedule (e.g., an integrated project schedule, a SE Master Schedule (SEMS)) based on objectives and work estimates.
 - – Determine the infrastructure and services needed for the project.
 - – Estimate the costs and establish a project budget.
 - – Plan the acquisition of materials, goods, and enabling systems.
 - – Generate and communicate a Systems Engineering Management Plan (SEMP), also called a Systems Engineering Plan (SEP), for project and technical management/execution, including the technical reviews and audits (see [Section 2.1.4](#)).
 - – Contribute to the quality management, configuration management, risk management, information management, and measurement plans to meet the needs of the project with regard to SE efforts (may be the SEMP for smaller projects).
 - – Establish the achievement criteria to be used for major milestones, decision gates, and internal reviews.
 - – Establish criteria for project performance.
- *Activate the project.*
 - – Obtain project authorization and resources. The Portfolio Management process provides this authorization (see [section 2.3.3.3](#)).
 - – Obtain authorization for the necessary project resources.
 - – Commence execution of the project plans.

Common approaches and tips:

- The SEMP (or equivalent technical planning) is an important outcome that identifies activities, key events, work packages, and resources. It references other planning artifacts that are tailored for use on the project.

- The standard ISO/IEC/IEEE 24748–4 on Systems Engineering Planning is a reference to aid in writing a SEMP.
- Plans for developing software are often captured in a Software Development Plan. (See ISO/IEC/IEEE 24748–5.)
- The creation of the Work Breakdown Structure (WBS) and other breakdown structures (e.g., Function Tree/Functional Breakdown Structure (FBS), Product Tree/Product Breakdown Structure (PBS), Organizational Breakdown Structure (OBS), Cost Breakdown Structure (CBS)) is an activity where SE and Project Management intersect (Forsberg, et al., 2005). (See paragraph on Breakdown Structures hereafter and [Section 5.3.3.](#))
- Taking shortcuts in the planning process reduces the effectiveness of other Technical Management Processes.
- Agile project management methods also include planning—the cycles may be shorter and more frequent, but planning is an essential process. Agile planning process is not related to the entire project but addressing only the next already known iterations while applying learning from the previous iterations.
- Defining project objectives, value, and the criteria for success are critical to guide project decision making. The project value should be expressed in technical performance measures (TPMs) (Roedler and Jones, 2006) (see [Section 2.3.4.7.](#)).
- Incorporate risk assessment early in the planning process to identify areas that need special attention or contingencies (see [Section 2.3.4.4.](#)). Always attend to the technical risks (PMI, 2013).
- If a Project Management Plan (PMP) already exists or is in preparation (in accordance with practices as defined by the Project Management Body of Knowledge (PMBOK®) (2021) from the Project Management Institute (PMI), for example), then it is important to coordinate in order to have a global consistency between these artifacts. The SEMP should reference, or provide a link to, the PMP for direction on how the SEMP will be updated and controlled on the project.

Elaboration

Project Planning Concepts. Project planning estimates the project budget and schedule against which project progress will be assessed and controlled. SE practitioners and PM practitioners must collaborate in project planning. SE practitioners perform technical management activities consistent with project objectives (see [Section 5.3.3](#)). Technical management activities include planning, scheduling, reviewing, and auditing the SE process as defined in the SEMP.

Systems Engineering Management Plan (SEMP). The SEMP is the key technical management plan that integrates the SE effort. It defines how the total set of engineering processes will be organized, structured, and conducted and how it will be controlled to provide a product that satisfies stakeholder requirements. The SEMP typically includes the following content (a complete outline can be found in ISO/IEC/IEEE 24748–4 (2016), which is aligned with ISO/IEC/IEEE 15288 and this handbook):

- organization of how SE interfaces with the other parts of the organization
- responsibilities and authority of the key engineering roles
- clear system boundaries and scope of the system
- key, technical objectives, assumptions, and constraints (or link to them)
- infrastructure support and resource management (i.e., facilities, tools, IT, personnel)
- technical schedule, including key milestones, decision gates, and associated criteria
- definition of the SE processes, including interaction with other engineering and project processes
- approach and methods for planning and executing the Technical Processes (see [Section 2.3.5](#))
- approach and methods for planning and executing the Technical Management Processes (see [Section 2.3.4](#))
- approach and methods for planning and executing applicable quality characteristic (QC) approaches (see [Section 3.1](#))
- major technical deliverables of the project

A SEMP should be prepared early in the project, submitted to the customer (or to management for in-house projects), and used in technical management for the concept and development stages of the project. The format of the SEMP can be tailored to fit project, customer, or company standards. In addition to being a stand-alone artifact, the SEMP can be a part of an integrated project plan, be a distributed set of plans, or be in a format other than a document (e.g., it may be composed of different models, management tools, or other artifacts).

The SEMS is an essential part of the SEMP and a tool for project control because it identifies the critical path of technical activities in the project. The schedule of tasks and dependencies helps prioritize the effort and justify requests for personnel and resources needed throughout the development life cycle.

Breakdown Structures. The purpose of the breakdown structures is to hierarchically decompose constructs in manageable and understandable elements. In projects, breakdown structures provide:

- a framework for ensuring that all requirements, functions, and products of the system design are identified and arranged in a logical relationship that can be traced to, and satisfy, the business and stakeholder needs;
- an identification of all activities and resources needed to the product;
- a cost relationship to the activities being performed;
- an organizational context for the project to perform the activities needed to the product;
- an identification, by name, within the organization of the responsible person for performing each activity;
- a basis for configuration control once a particular project breakdown structure is baselined, and a basis for effective management of changes;
- a framework to help identify risks and subsequent risk management;
- a basis for financial control and interface responsibilities resulting from business agreements.

The SE practice is to derive system functions from requirements and then allocate these functions into products or services, usually through the development of a functional and physical architecture (see [Section 2.3.5.4](#)). Functions and products are organized in breakdown structures that have the organizational framework of a tree, such as Function Tree and Product Tree. The Function Tree also, called Functional Breakdown Structure (FBS), is a breakdown of the functions of the required SoI into successively lower levels of its functional architecture. The Function Tree includes the technical characteristics of each function. The Product Tree, also called Product Breakdown Structure (PBS), is a breakdown of the SoI into successively lower-level details of its physical architecture (see [Section 1.3.5](#)).

The work to be carried out to reach the project objectives can be organized in a breakdown structure, as a hierarchical tree, where the lower-level activities provide more details. This is the Work Breakdown Structure (WBS), which is based on the FBS in the initial stages of system maturity (e.g., feasibility, conceptual design) and the PBS in the later stages. The WBS includes all activities needed to develop the product. Each branch of the WBS is used to define a work package (WP). Each WP describes the work to be performed, related input and output, who is doing the work, the related interfaces with other WPs, the related cost and deliverables and the key dates and milestones. The WBS serves as a reference for the identification of cost elements arranged into a Cost Breakdown Structure (CBS). Along with the overall management organization, the WBS is also used to determine who does what. This is represented in an Organizational Breakdown Structure (OBS), which is a hierarchical tree of the organizational elements. [Figure 2.24](#) illustrates these different project breakdown structures with their relationships. See [Section 5.3.3](#) for the relationship between PM and SE.

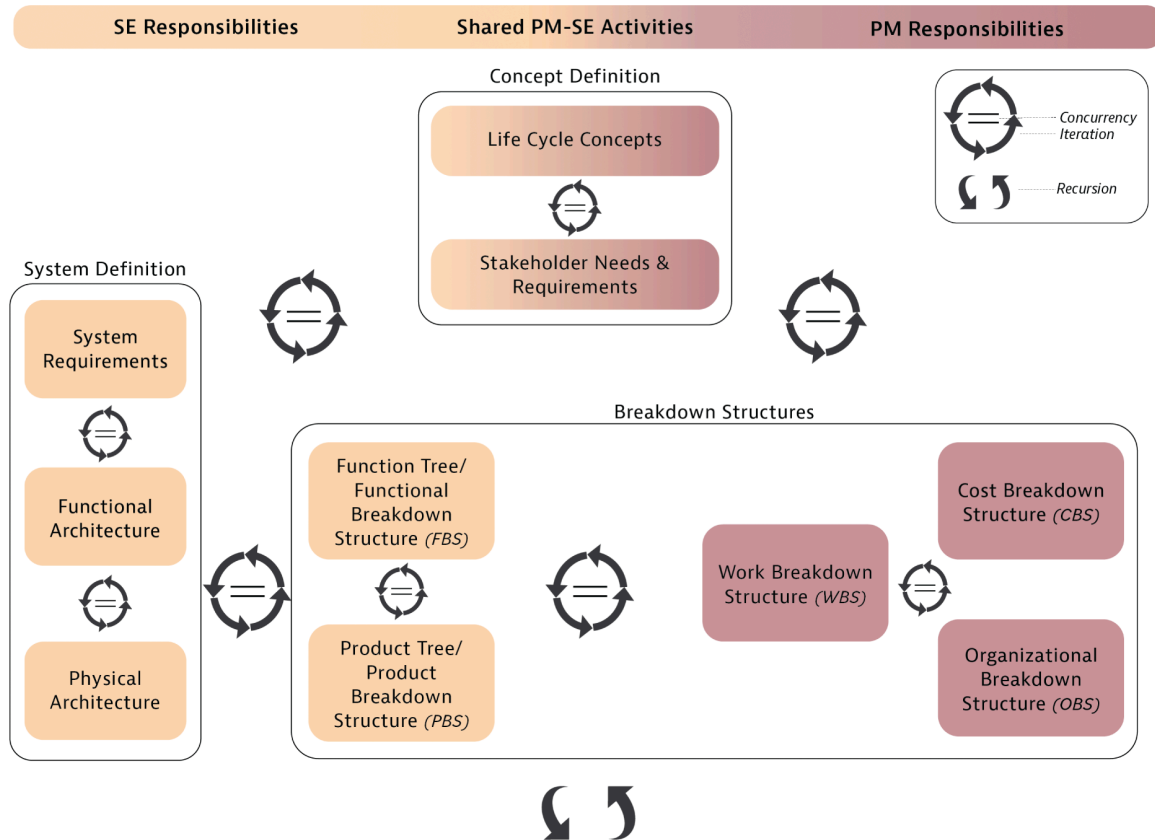


FIGURE 2.24 The breakdown structures. INCOSE SEH original figure created by Roussel and Dazzi on behalf of the INCOSE PM-SE Integration Working Group. Usage per the INCOSE Notices page. All other rights reserved.

2.3.4.2 Project Assessment and Control Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.3.2.1] The purpose of the Project Assessment and Control process is to assess if the plans are aligned and feasible; determine the status of the project, technical and process performance; and direct execution to help ensure that the performance is according to plans and schedules, within projected budgets, to satisfy technical objectives.

Assessments are scheduled periodically and for all milestones and decision gates. The intention is to maintain good communications within the project team and with the stakeholders, especially when deviations are encountered. The Project Assessment and Control process uses these

assessments to direct the efforts of the project, including redirecting the project when the project does not reflect the anticipated maturity.

Description The Project Planning process (see [Section 2.3.4.1](#)) identified details of the work effort and expected results. The Project Assessment and Control process collects data to evaluate the adequacy of the project infrastructure, the availability of necessary resources, and the compliance with project performance measures. Assessments also monitor the technical progress of the project and may identify new risks or areas that require additional investigation. A discussion of the creation and assessment of measures is found in [Section 2.3.4.7](#)—Measurement process.

The rigor of the Project Assessment and Control process is directly dependent on intrinsic characteristics of the project and the SoI, such as the complexity, urgency, and consequence of failure to deliver or failure of the SoI. Project control involves both preventive and corrective actions taken to ensure that the project is performing according to plans and schedules and within projected budgets. The Project Assessment and Control process may trigger activities within Technical Management Processes.

Inputs/Outputs Inputs and outputs for the Project Assessment and Control process are listed in [Figure 2.25](#). Descriptions of each input and output are provided in Appendix E.

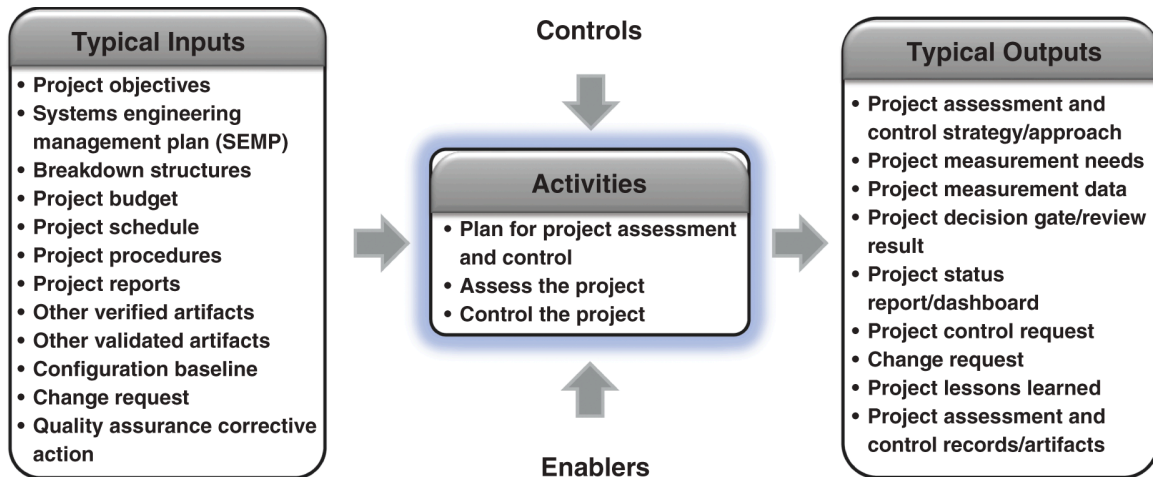


FIGURE 2.25 IPO diagram for Project Assessment and Control process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Project Assessment and Control process includes the following activities:

- *Plan for project assessment and control.*
 - – Develop a strategy/approach for assessment and control for the project.
- *Assess the project.*
 - – Determine whether project objectives and plans are aligned with the project context.
 - – Determine cost, schedule, and performance variances for the project and technical effort through assessment of status versus plans.
 - – Evaluate the effectiveness and efficiency of the performance of project activities.
 - – Determine if the project roles, responsibilities, accountabilities, and authorities are adequate.
 - – Assess the adequacy and the availability of the project infrastructure and resources.
 - – Using project measures and milestone status, assess the progress of the project.
 - – Conduct required reviews, audits, and inspections to determine readiness to proceed to the next milestone (see paragraph on reviews and audits in the elaboration and [Section 2.1.4](#) for more details).

- – Monitor areas of high risk, such as critical tasks and new technologies/applications (see [Section 2.3.4.4](#)).
- – Recommend actions that are warranted by measurement results and other project information.
- – Make recommendations for adjustments to project plans—these are input to the project control process and other decision-making processes.
- – Record and provide the project status report, gathering all needed measures on technical progress aspects (e.g., performance, requirements compliance, verification and validation progress) and technical management aspects (e.g., schedule, cost, risk, configuration status).
- – Communicate status in the project status report as designated in agreements, policies, and procedures.
- *Control the project.*
 - – Initiate preventive actions when assessments indicate a trend toward deviation.
 - – Initiate problem resolution when assessments indicate nonconformance with performance success criteria.
 - – Initiate corrective actions when assessments indicate deviation from approved plans.
 - – Update project planning as needed based on the project control or corrective actions.
 - – Implement change actions to reflect contractual changes to cost, time, or quality. This is usually due to the impact of acquirer or supplier request.
 - – Authorize the project to proceed when assessments support a decision gate or milestone event.

Common approaches and tips:

- One way to remain updated on project status is to conduct regular team meetings. Short stand-up meetings on a daily or weekly schedule are effective for smaller groups.

- Prevailing wisdom suggests that “what gets measured gets done,” but projects should avoid the collection of measures that are not used in decision making.
- Good practices show that status should be concise and visual (e.g., usage of Red/Yellow/Green “traffic lights”) in order to quickly and easily identify the critical issues on which urgent actions for recovery are required. Another useful tool is a project dashboard that provides a timely and easy summary of status.
- A template for the project status report is a good practice. This template may be included in the SEMP (or PMP).
- Methods and techniques for Project Assessment and Control should be formally described in the SEMP (or PMP) and agreed with the project team.
- The Project Management Institute (PMI) provides industry-wide guidelines for project assessment, including Earned Value Management techniques.
- Project teams need to identify critical areas and control them through measurement, risk management, analysis, configuration management, and information management.
- The Project Assessment and Control process requires close cooperation between the PM practitioner and SE practitioner, with PM being accountable for the overall results of the project and SE being accountable for the achievement of the technical activities.
- The typical common responsibilities between PM and SE practitioners are risk management, external supplier relations, quality management and life cycle planning.
- An effective feedback control process is an essential element to enable the improvement of project performance.
- Incremental and evolutionary models typically schedule frequent assessments and make project control adjustments on tighter feedback cycles than sequential development models (see [Section 2.2](#)).
- Tailoring of organization processes and procedures (see [Section 4.1](#)) should not jeopardize any certifications. Processes must be established with effective reviews, assessments, audits, and improvements.

- Standard ISO/IEC 24748–8 / IEEE 15288.2 (2014) is a useful reference on how to define and manage technical reviews and establish requirements for the related milestones.

Elaboration

Integration of Technical Management Artifacts. Each of the Technical Management Processes provides essential insight into the health and progress of the project through the life cycle with respect to the specific focus of the particular process. However, it is important to look at the results of these processes in an integrated view, especially since there are relationships between these processes and their artifacts. For example, the results of the Measurement process provide useful insights into risks, technical reviews and audits, and quality assurance, as well as many other things. Similarly, other processes may identify new information needs for which new measures should be initiated. Mechanisms should be put in place to provide an integrated view of the results or artifacts in a way that the decision makers can interpret quickly and see trends and trigger points to aid decisions. Two such mechanisms are the project status report and the project dashboard. Both organize and provide a summary of similar information about the project; the status report usually presents the information in report form and the dashboard is usually a digital representation that uses gauges, graphs, indicators, or other visual representations of the information. In both mechanisms relationships are shown and trends or areas needing attention are highlighted.

Technical Reviews and Audits. Technical reviews and audits are a foundational element of an effective SE approach and form the backbone of robust technical assessment. Technical reviews and audits provide a venue for baselining stakeholder and system requirements, evaluating the system's technical maturity, and identifying and assessing risks to system performance, cost, and schedule. In order for a project's technical management to have a balanced information basis on which to base any required project control actions, each technical review or audit should be conducted from an integrated project viewpoint, including technical status

and progress, cost and schedule status, and impacts and risk assessment, to help ensure that technical review decisions do not create unrecognized and unacceptable future project impacts. See [Section 2.1.4](#) Technical Reviews and Audits and ISO/IEC 24748–8 / IEEE 15288.2 (2014) for more information.

2.3.4.3 Decision Management Process

Overview

Purpose As defined by ISO/IEC/IEEE 15288,

[6.3.3.1] The purpose of the Decision Management process is to provide a structured, analytical framework for objectively identifying, characterizing and evaluating a set of alternatives for a decision at any point in the life cycle and select the most beneficial course of action.

[Table 2.4](#) provides a partial list of decision situations (opportunities) that are commonly encountered throughout a system's life cycle. Buede and Miller (2009) provide a much larger list.

TABLE 2.4 Partial list of decision situations (opportunities) throughout the life cycle

Life cycle stage	Decision situation (opportunity)
Concept	Assess technology opportunity/initial business case Craft a technology development strategy Inform, generate, and refine a capability artifact Conduct analysis of alternatives Supporting program initiation decision Select system architecture
Development	Select system element Select lower-level elements Select verification and validation methods Perform make-or-buy decision
Production	Select production process and location
Utilization, support	Select maintenance approach
Retirement	Select disposal approach
INCOSE SEH original table created by Parnell, Kenley, and Roedler. Usage per the INCOSE Notices page. All other rights reserved.	

Decision management as a critical SE activity. Consider the number of decisions involved in identifying a business/mission need, crafting a technology development strategy, defining the stakeholder and system requirements, selecting a system architecture, converging on a detailed design, developing verification and validation plans, determining make-or-buy decisions, creating production ramp-up plans, crafting maintenance and logistics plans, and selecting disposal approaches. New product

developments entail an array of interrelated decisions throughout the system life cycle.

Description The Decision Management process transforms a broadly stated decision situation into a recommended course of action and associated implementation plan. The process requires a decision maker with full responsibility, authority, and accountability for the decision, a decision analyst with a suite of decision tools, subject matter experts with performance models, and a representative set of end users and other stakeholders (Parnell, et al., 2013). The decision process is executed within the policy and guidelines established by the system sponsor. A well-structured decision process will capture and communicate the impact that different value judgments have on the overall decisions and facilitate the search for alternatives that remain attractive across a wide range of value schemes.

Inputs/Outputs Inputs and outputs for the Decision Management process are listed in [Figure 2.26](#). Descriptions of each input and output are provided in Appendix E.

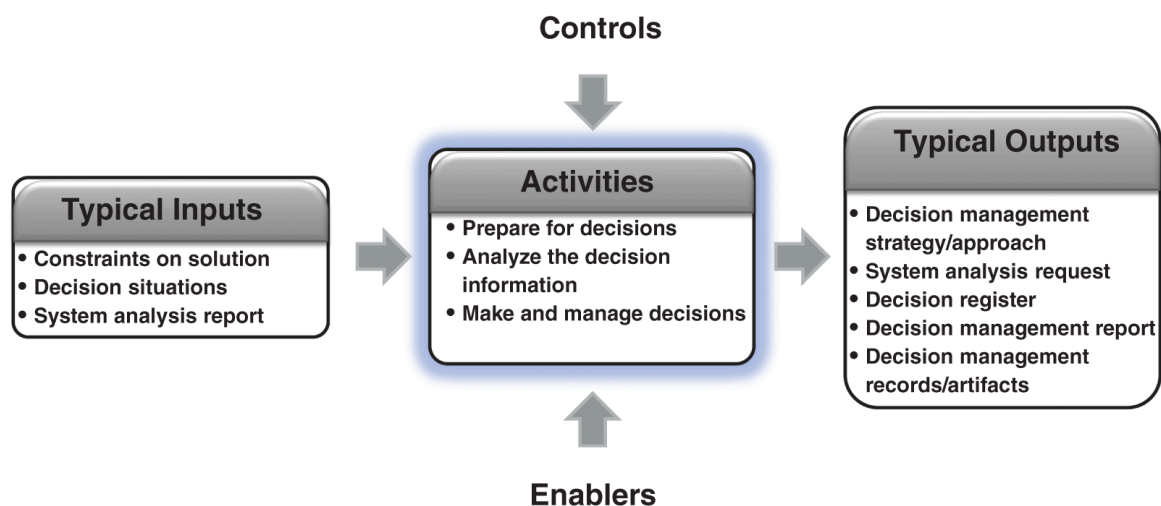


FIGURE 2.26 IPO diagram for the Decision Management process. INCOSE SEH original figure created by Walden, Shortell, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Decision Management process includes the following activities:

- *Prepare for decisions.*

- – Develop the decision management strategy/approach for system or project decisions.
- – Establish and challenge the decision statement and clarify the decision to be made.
- – Determine the analyses methods, other processes, and tools required to support decision activities. (Note that the System Analysis process (see [Section 2.3.5.6](#)) is often applied to perform analyses to provide input for the decisions.)
- – Provide resources to implement the strategy.
- *Analyze the decision information.*
 - – Frame, tailor, and structure each decision.
 - – Develop objectives and measures.
 - – Generate creative alternatives.
 - – Assess alternatives via deterministic analysis.
 - – Synthesize results.
 - – Identify uncertainties and conduct probabilistic analysis.
 - – Assess impact of the uncertainties.
 - – Improve alternatives.
 - – Communicate trade-offs.
 - – Present recommendation and implement action plan.
- *Make and manage decisions.*
 - – Record the decision with relevant data, models, and supporting documentation (i.e., the decision authority, source, and rationale)
 - – Describe analyses methods, other processes, and tools actually used to support decision activities.
 - – Communicate new directions from the decision

Common Approaches and Tips:

- Since there are many decisions across the spectrum of project management, system definition, and life cycle activities, the Decision Management process is applied in conjunction with most of the Technical Management and Technical Processes.
- It is important to verify and validate the data and assumptions used in the decision analyses, since the validity of the analysis results depends

on the use of valid data and assumptions, and the application of appropriate analytic methods.

Elaboration

SE practitioners face many decision situations throughout the life cycle of a project. They must choose the analytical approach that best fits the frame and structure of each decision problem. For instance, when there are “clear, important, and discrete events that stand between the implementation of the alternatives and the eventual consequences” (Edwards, et al., 2007), a decision tree is often a well-suited analytical approach, especially when the decision structure has only a few decision nodes and chance nodes. As the number of decision nodes and chance nodes grows, the decision tree quickly becomes unwieldy and loses some of its communicative power. Furthermore, decision trees require end node consequences be expressed in terms of a single number.

The decision management method most employed by SE practitioners is a multiple objective decision approach (MODA) (Parnell, 2016), in which an objective function is formulated to synthesize an alternative’s response across multiple, often competing, objectives. The aim is to define, measure, and assess stakeholder value and then synthesize this information to facilitate the decision maker’s search for alternatives that represent the best balance with often competing objectives. If time and funding allow, SE practitioners may want to conduct trade studies using several techniques, compare results, and reconcile any differences to ensure findings are robust.

The following are a summary of decision management good practices.

Framing, Tailoring, and Structuring Decisions. Capturing a description of the system baseline, as well as the concept of operations with some indication of system boundaries and anticipated interfaces, helps ensure the understanding of the decision context. This includes such details as the time frame allotted for the decisions, an explicit list of stakeholders, a

discussion regarding available resources, and expectations regarding the type of action to be taken as a result of the decision at hand. It may also include decisions anticipated in the future (Edwards, et al., 2007).

Developing Objectives and Measures. Defining the decision to be made may require balancing a large number of ambiguous and potentially conflicting stakeholder need statements, engaging in uncomfortable discussions regarding the relative priority of each requirement, and establishing walkaway points and stretch goals. Per Keeney (2002):

“Most important decisions involve multiple objectives, and usually with multiple-objective decisions, you cannot have it all. You will have to accept less achievement in terms of some objectives to achieve more on other objectives. But how much less would you accept to achieve how much more?”

Use the information obtained from the Business or Mission Analysis, Stakeholder Needs and Requirements Definition, System Requirements Definition, System Architecture Definition, and Design Definition processes to develop objectives and measures for MODA models that use fundamental objectives (why, what, where, and when), but not means objectives (how). For each fundamental objective, a measure must be established so that alternatives that more fully satisfy the objective receive a better score on the measure than those alternatives that satisfy the objective to a lesser degree. These measures (also known as measures of effectiveness (MOEs), key performance parameters (KPPs), measures of performance (MOPs), technical performance measures (TPMs), critical performance measures, attributes, criterion, or metrics) must be unambiguous, comprehensive, direct, operational, and understandable (Keeney and Gregory, 2005) (Roedler and Jones, 2005) (see [Section 2.3.4.7](#)).

Generating Creative Alternatives. For many trade studies, the alternatives will be systems composed of many interrelated system elements. It is important to establish a meaningful product structure for the SoI and to apply this product structure consistently throughout the decision analysis.

The product structure should be a useful decomposition of the elements of the SoI that explores the trade space. Each alternative is composed of specific design choices for each element. The ability to communicate the differentiating design features of the alternatives is essential. An alternative to a finite number of alternatives is Set-Based Design (SBD). SBD has been shown to effectively and efficiently explore the trade space (Specking, et al., 2018).

Assessing Alternatives via Deterministic Analysis. The decision team should engage subject matter experts by creating models using operational and test data along with the defined objectives, measures, and alternatives to assess performance and using structured scoring sheets. Each score sheet contains a summary description of the alternative and the scoring criteria. Ideally, the models and simulations should be integrated with the performance, value, and cost models so a design change impacts all models.

Synthesizing Results. Using the data summarized in the objective measure consequence table, explore, understand, aggregate the data, and display results in a way that facilitates stakeholder understanding.

Identifying Uncertainty and Conducting Probabilistic Analysis. It is important to identify potential uncertainty surrounding the assessed score and variables that could impact one or more scores (see [Section 1.4.1](#)). One example of uncertainty is that system concepts are described as a collection of system element design choices, but knowledge of the system element performance during system design is often incomplete. Subject matter experts can often assess an upper, nominal, and lower bound score by making three separate assessments: (i) assuming a low performance, (ii) assuming moderate performance, and (iii) assuming high performance.

Assessing Impact of Uncertainty. Decision analysis uses many forms of sensitivity analysis including line diagrams, tornado diagrams, waterfall diagrams, and several uncertainty analyses, including Monte Carlo simulation, decision trees, and influence diagrams (Parnell, et al., 2013). Monte Carlo simulations are used to identify the relative impact of each

source of uncertainty on the performance, value, and cost of each alternative. Risks should be identified when significant uncertainty is present.

Improving Alternatives. One could be tempted to end the decision analysis here, highlight the alternative that has the highest total value, and claim success. Such a premature ending would not be considered good practice. Good practice includes further analysis to mine the data generated for the first set of alternatives to reveal opportunities to modify some system element design choices to identify untapped value and reduce risk.

Communicating Trade-Offs. The decision team should identify key observations regarding what stakeholders seem to want and what they may be willing to give up to achieve it. The decision team highlights the design decisions that are least significant and/or most influential and provide the best stakeholder value. In addition, the important uncertainties and risks should also be identified. Observations regarding combinatorial effects of various design decisions are also important products of this process step. Finally, competing objectives that are driving the trade-offs should be highlighted as well.

Presenting Recommendations and Implementing the Action Plan. It is helpful to clearly describe the recommendation as an actionable task list to increase the likelihood of the decision analysis leading to some form of action showing tangible value. Decisions should be documented using digital engineering artifacts. Reports that include the analysis, decisions, and rationale are important for historical traceability and future decisions.

2.3.4.4 Risk Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.3.4.1] The purpose of the Risk Management process is to identify, analyze, treat and monitor the risks continually.

Description Risk Management is a disciplined approach to dealing with the uncertainty that is present throughout the entire system life cycle (see [Section 1.4.1](#)). Opportunity management may be performed in conjunction with or as part of risk management. A primary objective of risk management is to identify and manage uncertainties that threaten or reduce the value provided by a business enterprise or organization. A primary objective of opportunity management is to identify and manage uncertainties that enhance or increase the value provided by a business enterprise or organization. Since risk cannot be reduced to zero, another objective is to achieve a proper balance between risk and opportunity.

Risk management, as it relates to SE, is defined in ISO/IEC/IEEE 15288 and elaborated upon in ISO/IEC/IEEE 16085 (2021). As stated in ISO/IEC/IEEE 16085,

[6.1] The Risk Management process is a continual process for systematically addressing risk throughout the life cycle of a system, product, or service. It can be applied to risks related to the acquisition, development, maintenance, or operation of a system.

When using this process for opportunity management, the above statement, with the term “opportunity” substituted for the term “risk,” is also true.

Inputs/Outputs Inputs and outputs for the Risk Management process are listed in [Figure 2.27](#). Descriptions of each input and output are provided in Appendix E.

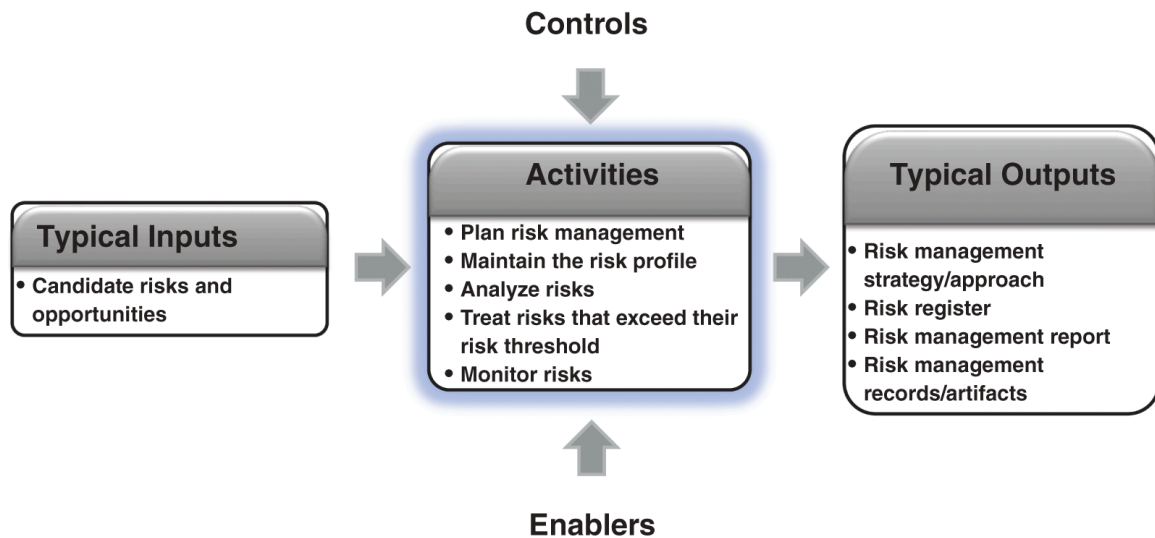


FIGURE 2.27 IPO diagram for Risk Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Risk Management process includes the following activities:

- *Plan risk management.*
 - – Develop the risk management strategy/approach.
 - – Capture the Risk Management process context, including risk categories.
- *Maintain the risk profile.*
 - – Capture the thresholds and conditions of the risks.
 - – Establish and maintain a risk profile to include context of the risk and its likelihood of occurrence, severity of consequences, risk thresholds, and priority and the risk action requests along with the status of their treatment.
 - – Ensure updates of the risk profile are available to relevant stakeholders.
- *Analyze risks.*
 - – Identify risks consistent with the risk management strategy/approach.
 - – For each risk, estimate its likelihood and consequence of occurrence.
 - – For each risk, use the risk thresholds to evaluate the risk for potential treatment.

- – For risks that exceed the threshold, capture recommended treatment strategies and measures.
- *Treat risks that exceed their risk threshold.*
 - – Identify a set of feasible alternatives for the treatment of risks.
 - – Establish measures to provide insight into the risk treatment effectiveness.
 - – Execute the treatments for the risks.
 - – When management action is needed for risk treatments, ensure effective coordination.
- *Monitor risks.*
 - – Maintain the record of risk items and how they were treated.
 - – Monitor high priority risks.
 - – Monitor risks and the risk management context to capture changes and update priorities and actions.
 - – Throughout the life cycle, monitor for new risks and sources of risk.

When using this process for opportunity management, the above process activity description, with a few adjustments in terminology, and the term “opportunity” substituted for the term “risk,” is reasonable.

Common approaches and tips:

- In the Project Planning process, a risk management plan (RMP) is tailored to satisfy the policies, procedures, standards, and regulations related to and affecting the management of risks for the project.
- Process Enablers—It has been found that an organization’s structure and culture can have a significant effect on the performance of the Risk Management process. ISO 31000 (2018), outlines a model that advocates the establishment of principles for managing risk and a framework for managing risk that work in concert with the process for managing risk.
- Typical strategies for coping with risk include transference, avoidance, acceptance, or taking action to reduce the potential negative effects of the situation.
- Most Risk Management processes include a prioritization scheme whereby risks with the greatest potential negative consequence and the

highest likelihood are treated before those deemed to have lower potential negative consequences and lower likelihood. The objective of risk management is to balance the allocation of resources such that a minimum amount of resources achieves the greatest risk mitigation (or opportunity realization) benefits.

- Communication errors and misunderstandings can be prevented by defining and communicating the risk terminology to be used by the project and including with the project's risk management plan (RMP).
- Experience has shown that terms such as "positive risk" and concept models that define opportunity as a subset of risks serve only to confuse. Take care to define the terminology and concepts to be used by the project team and provide training to reinforce a common understanding.
- Practices used for writing good requirements help with risk statements. For example, one good practice for identifying and clarifying risks is to use an "if <situation>, then <consequence>, for <stakeholder >" pattern. This pattern helps to determine the validity of a risk and assess its magnitude or importance.
- Risk management is most successful when risk-based thinking is embraced and integrated into the culture. All personnel are responsible for identifying risks early and continuously throughout the project life cycle.
- Negative feedback toward personnel who identify a potential problem will discourage the full cooperation of engaged stakeholders and could result in failure to identify and address serious risk-laden situations. Conduct a transparent Risk Management process to encourage all stakeholders to assist in risk mitigation efforts.
- Some situations can be difficult to categorize in terms of probability and consequences; involve all relevant stakeholders in this evaluation to capture the maximum variety in viewpoints.
- Risk measurement is not an exact science. Variation in stakeholder perspectives, perceptions, and tolerance levels, along with high uncertainty in available data, can make reliance on quantitative measures of risk insufficient. For example, some low-likelihood/high-

severity risks might require treatment and monitoring regardless of the estimated likelihood of occurrence (Taleb, 2018) (Siegel, 2019).

- External risks are often neglected in project management. External risks are risks caused by or originating from the surrounding environment of the project (Fossnes, 2005). Project participants often have no control or influence over external risk factors, but they can learn to observe the external environment and eventually take proactive steps to minimize the impact of external risks on the project. The typical issues are time-dependent processes, rigid sequence of activities, one dominant path for success, and little slack.

Elaboration

Definitions of Risk. Few terms used in engineering have as many different published definitions as the term “risk.” In practice, risk terminology and concepts vary considerably across industries; however, most published definitions of risk align with one of two concept models. Below are two prominent definitions of risk that capture the essence of both concepts:

- The effect of uncertainty on objectives [see ISO/IEC/IEEE 15288, ISO Guide 73, ISO/IEC/IEEE 16085, ISO/IEC 31000, ISO 27000]
- The combination of the probability of occurrence of harm and the severity of that harm [see ISO Guide 51, ISO 22367, ISO 14971]

Both definitions may be used in an SE project. The first definition includes the concept that effects may be negative or positive. In this respect the first definition accommodates use of the second definition. In SE it is common to use the term “risk” when referring to scenarios with a negative effect, and the term “opportunity” when referring to scenarios with a positive effect. The second definition (which accommodates only negative effects) is commonly used in safety engineering, and its use may be required in order to demonstrate compliance to risk management standards and regulations applicable to products and systems that impact public health, safety, and security. For example, in the medical industry (see [Section 4.4.2](#)), particularly for medical devices, risk management is often centered on